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Evaluating low-carbon mechanisms for hydrogen penetration in Iran's energy supply system

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ABSTRACT

Hydrogen, as a clean fuel, can be a suitable replacement for conventional fossil fuels and can emerge as an energy carrier. Iran, with its abundant fossil fuel resources and high capacities in renewable energies, has a good opportunity to produce and export hydrogen in the coming decades. To increase the penetration of hydrogen in the country's energy mix, it is necessary to integrate this clean fuel into the energy supply system. Therefore, there's a need to evaluate the hydrogen integration/penetration and its potential impact within the energy mix along with other energy carriers. Several low-carbon financial mechanisms may be planned and applied to incentivize this transition. In this regard, the present research has modeled and optimized Iran's energy supply system with a focus on hydrogen penetration under low-carbon strategies such as imposing carbon taxes, low-rate facilities to the hydrogen supply chain, imposing caps on greenhouse gas (GHG) emissions, and eliminating energy subsidies over the period 2020 to 2055 using an Open-Source Energy Modeling System (OSeMOSYS) framework. Results indicate that by following the current trend and investing within the range of US$12-15 billion, there is the potential to construct 24 to 32 GW of new capacity for hydrogen production in the country. Subsequently, hydrogen production pathways in each scenario are examined, and economic and environmental performance indicators are used to evaluate and compare the mechanisms. The outcomes of this study provide a comprehensive roadmap for expanding hydrogen production and and understanding the impact of this energy carrier on other parts of the energy supply system in a fossil fuel rich country like Iran.

Nomenclature

GHG	Greenhouse gas
OSeMOSYS	Open-source energy modeling system
MCDA	Multi-criteria decision analysis
LES	Linear Expenditure System
FCEV	Fuel cell electric vehicles
MENA	Middle East and North Africa
LNG	Liquified natural gas
SCI	Statistical center of Iran
IFCO	Iranian fuel conservation company
CBI	Central Bank of Iran's
CC PP	Combined cycle power plants
SMR	Steam methane reforming
CCS	Carbon capture and storage
CRF	Capital recovery factor
INDC	Intended nationally determined contribution
OCGT	Open cycle gas turbine

1. Introduction

As the global population rises, energy demand increases substantially due to its necessity across numerous sectors such as industry, work, cleaning, transportation, and commuting [1]. On the other hand, energy consumption has significantly outgrown economic output, resulting in an upward trend in energy intensity, contrasting the downward trend seen in most developed countries [2,3]. Addressing future energy demand while reaching net-zero carbon emissions requires a decarbonized power system with a significant focus on minimizing emissions from these sectors [4,5]. To achieve that, a transition from direct fossil fuel use towards low-carbon energy sources such as renewables [6], green/blue hydrogen [7], and bioenergy [8], and nuclear [9] is required, followed by enhancements in efficiency and a decrease in overall energy consumption [10]. As an energy carrier, hydrogen stands out as a promising alternative to traditional fossil fuels, with its potential for production from water, unique consumption characteristics, and significant environmental benefits. Hydrogen also

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holds potential as a solution for increasing the share of renewable energy usage and storage within energy systems [11–13]. However, challenges arise with alternative approaches, therefore, developing a comprehensive energy system model including the prospective hydrogen supply and demand into it is recommended. This integration with a detailed focus on aspects such as potential technology adoption, policy incentives, and market dynamics contributes to evaluating energy system dynamics and formulating policies for fostering sustainable energy development [14–16].

1.1. Energy supply system in Iran

Iran was the fifth-largest crude oil producer and the fourth-largest natural gas producer globally in 2021 [17]. Additionally, it holds the second-largest natural gas reserves and the third-largest oil reserves in the world. The country accounts for 24% of the Middle East's oil reserves and 12% of the world's total [18]. On the other hand, fossil fuel energy carriers constitute the majority of Iran's energy consumption. According to Fig. 1, over 90% of Iran's total primary energy consumption demand is fulfilled via gas and oil.

On the other hand, Iran has been ranked among the top ten greenhouse gas emitters in recent studies [20]. This ranking is attributed to the nation's energy-intensive economic growth and carbon-intensive portfolio. Fossil fuel combustion is considered a significant anthropogenic parameter contributing to climate change and a primary cause of these emissions [21]. To increase its energy supply security, Iran has shown a growing interest in expanding its energy mix [22]. According to data from the Renewable Energy and Energy Efficiency Organization of Iran (SATBA), Iran's renewable energy capacity, including solar, wind, hydro, and biomass, is increasing, although it still represents a small portion of the overall energy mix. The total installed capacity of renewable energy in Iran will reach approximately 850 MW by 2023, marking a small but significant shift towards cleaner energy sources [23].

These issues highlight the importance of energy modeling in Iran, especially in terms of optimizing energy production, reducing GHG emissions, and transitioning towards a more sustainable energy system. In this regard, research and studies on energy modeling in Iran have been conducted to effectively plan and implement energy efficiency strategies, demand management, and integration of renewable resources, as detailed in Table 1. These models can facilitate a systematic assessment of the impacts of various energy policies, investment decisions, and the potential of renewable energy under different economic and technological scenarios. Some studies have utilized conventional energy supply system models such as TIMES and MESSAGE, while others have developed a model particularly for their research objective.

1.2. Hydrogen supply chain modeling

The transition towards a hydrogen-based economy requires a deep understanding of both supply and demand sides of this energy carrier, where energy system modeling plays a vital role [15]. Modeling the supply chain requires a comprehensive analysis of resource availability, technology efficiency, infrastructure needs, and environmental impacts [34]. On the other hand, hydrogen demand is diverse, encompassing sectors such as transportation, where hydrogen fuel cells replace battery electric vehicles, to industrial applications including ammonia and steel production. The energy sector also sees hydrogen as a potential game-changer for energy storage and generation. Modeling hydrogen demand depends on factors such as forecasting consumption patterns based on technology adoption, policy incentives, market dynamics, and how it integrates into existing energy systems [16]. Table 2 presents a summary of research that assessed hydrogen's role in using energy system modeling.

In addition to the above, in many studies, the feasibility of hydrogen production through specific technology has been evaluated. Therefore, hydrogen emerges as a promising solution in future energy systems, particularly due to its capacity for storing excess electricity from renewable energy sources (e.g., solar and wind) while also fueling emission-free fuel cell electric vehicles (FCEV) [43]. Following the similar context, in a study conducted by Meng et al., efficient photocatalytic systems were developed for hydrogen production via ammonia borane hydrolysis, demonstrating high hydrogen evolution rates under mild conditions [44], and the research progress on various metal sulfide photocatalysts in photocatalytic hydrogen evolution using multilevel biomass resources and waste plastics is reviewed by Ban et al. [45]. The use of catalysts highlights progress toward sustainable and clean hydrogen technologies. Also, to meet 2045 carbon neutrality targets, a study investigated hydrogen demand scenarios in California's transportation sector, resulting in detailed infrastructure requirements,

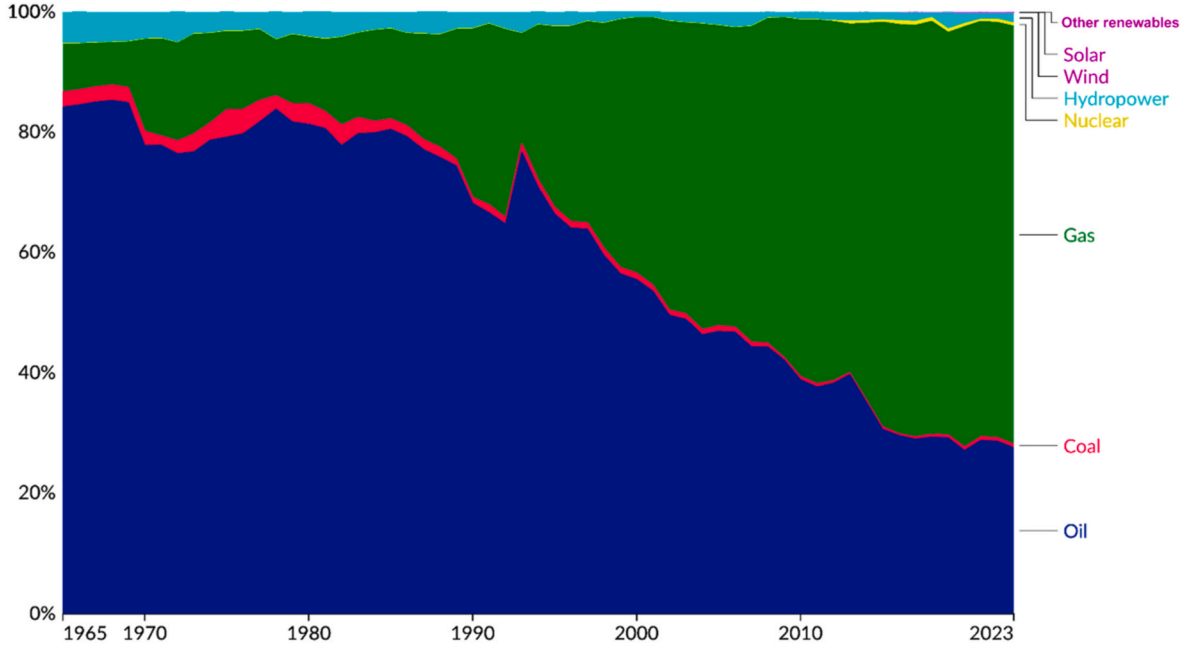


Fig. 1. The Iran energy mix by source [19].

Table 1
Selected energy system modeling and optimization studies in Iran.

Document	Research aims	Methodology	Ref.
Sharifian et al., 2024	Modeling Iran's energy supply system to reduce GHG and focus on the electricity generation sector	"OSeMOSYS" model	[24]
Zamanipour et al., 2023	Investigating the Impacts of Climate Change and Temperature Rise on Electricity Demand within the Energy Supply System	"TIMES" model	[25]
Aryanpur et al., 2019	Modeling and optimization to achieve a low-carbon electricity transition	"MESSAGE" model	[26]
Ghadaksaz et al., 2020	Energy Supply System Modeling for Greenhouse Gas Reduction, to Achieve Paris Agreement Commitments	"MESSAGEix" model	[14]
Ahmadi et al., 2023	Improving Renewable Energy Capacity within the Electricity Supply Systems	EnergyPlan "software"	[27]
Noorollahi et al., 2021	Carbon Dioxide Emission Reduction and Energy Transition in achieving a sustainable energy system	EnergyPlan "software"	[28]
Ahmadi et al., 2020	Mitigating Environmental Pollutants in the Domestic Sector, by estimating optimal useful energy use	a bottom-up algorithm	[29]
Eshraghi et al., 2016	Modeling Energy Transition Paths for Oil Refineries and Power Plants Considering New Technologies	"RAISE" model	[30]
Sahabmanesh et al., 2017	predicting useful energy demand and energy systems model's sustainability of Hamedan	Multi-criteria Decision Making Analysis (MCDA)	[31]
Aryanpur et al., 2015	A carbon tax cost optimization model of electricity supply system to investigate the integration of renewable sources	"MESSAGE" model	[32]
Barkhordar et al., 2014	Modeling useful energy demand in rural and urban sectors	Linear Expenditure System (LES) model	[33]

technology and cost insights along the hydrogen supply chain, and analysis of key factors influencing end-user hydrogen costs [46]. Similarly, Li et al. showcased the domestic focus of current hydrogen supply chain network design models and emphasized the potential benefits of international hydrogen trade, while acknowledging potential challenges associated with long-distance transportation and cross-national financing activities [47].

Also, in recent years, the development of hydrogen production technologies has gained attention in Iran and MENA (Middle East and North Africa) countries, where fossil fuels play a significant role in their energy systems [48]. Rahimirad and Sadabadi have recommended policy strategies for the development of green hydrogen in Iran [49]. Moreover, Qolipour et al. conducted a techno-economic feasibility study for the development of a photovoltaic-wind power plant to produce electricity and hydrogen in the Hendijan region of southwestern Iran [50] and Mohebbi et al. developed an optimization model for the development of a hydrogen port in the Khuzestan province of Iran [51].

In another study, Hassan et al. examined the potential and barriers to hydrogen production in Saudi Arabia in line with the country's energy system [52]. Also, a comprehensive roadmap for the potential production, use, and export of carbon-free hydrogen in Qatar was investigated by Okonkwo et al. [53], and Zaiter et al. conducted a feasibility study of using hydrogen as a renewable energy carrier in industry energy demand in the United Arab Emirates [54]. Furthermore, a technical, economic, and environmental feasibility study for hydrogen production from wind and solar electricity in the MENA region was conducted by Nasser and Hassan. They demonstrated each country's potential for hydrogen production and concluded that establishing hydrogen production infrastructure could create a tremendous opportunity in this region [55].

Many Middle Eastern countries have national and long-term plans to

Table 2
Research focuses on the modeling of the energy system along with the hydrogen supply chain.

Document	Country or Region	Research aims	Methodology	Ref.
Shirizadeh et al., 2023	France	Modeling and optimization of hydrogen penetration produced from electricity under different scenarios	Linear programming model	[7]
Husarek et al., 2021	Germany	Energy system model with renewable and nonrenewable resources for the hydrogen supply chain	Linear programming model	[35]
Koutsandreas et al., 2023	Greece	Modeling the electricity supply system to increase green hydrogen production in the country	"OSeMOSYS" model	[36]
Navas-Anguita et al., 2020	Spain	Technical-economic modeling with environmental approaches of the energy supply system along with the hydrogen supply Chain	LEAP-OSeMOSYS	[37]
Seck et al., 2022	Europe	The potential role of low-carbon and renewable hydrogen in achieving Europe's emission reduction targets with three complementary models and assesses two policy-driven scenarios focusing on technology diversification and renewable energy advances.	A bottom-up model based on the TIMES	[38]
Al-Khelaiwi et al., 2024	Saudi Arabia	Technical and economic feasibility study for green and blue hydrogen production	Linear programming model	[39]
Li et al., 2020	China	a long-period, multi-region, and monthly-scale optimization model for the hydrogen supply chain	Mixed integer linear programming model	[40]
Plazas-Niño et al., 2024	Colombia	techno-economic assessment of hydrogen pathways and their cross-sectoral interactions within the national energy system	"OSeMOSYS" model	[41]
Vargas-Ferrer et al., 2023	Chile	Evaluation of the power supply system for the integration of electrolyzer technology for hydrogen production	"OSeMOSYS" model	[42]

develop new infrastructure and technologies to maximize hydrogen production. In recent years, Saudi Arabia, Oman, and the United Arab Emirates can be named as leaders in this matter. These countries' focus on increasing hydrogen production, especially from renewable sources, has attracted the attention of many countries for investment. In this regard, the 2 GW Neom electrolyzer project in Saudi Arabia can be mentioned [56]. Additionally, Oman is also working to build the largest green hydrogen plant in the Arabian Sea by 2038 [57].

1.3. Research gaps and novelty

Reviewing the literature reveals that following the transition towards an eco-friendly energy system, introducing hydrogen as a clean energy carrier becomes more promising, with a significant increase in demand anticipated for the future decades. Unlike the global interest in hydrogen, the specifications for integrating it into Iran's energy system model have not been appropriately addressed. Most countries, however, have developed detailed long-term plans for hydrogen production and use [58–60]. Given their availability, competitive pricing, ease of storage and distribution, and high hydrogen-to-carbon ratio, fossil fuels are likely to remain the main source of hydrogen production in the short and medium term. Therefore, to ensure a comprehensive and valid analysis of Iran's energy system model and effectively predict future supply and demand, hydrogen should be integrated and modeled as an energy carrier within this system. To achieve this, Iran's energy supply system was first modeled using OSeMOSYS, with the results evaluated and validated against actual data and previous studies. Following this, the entire hydrogen supply chain, along with other technologies, was included in the model to identify opportunities for expanding the hydrogen industry within Iran's energy system. To bridge existing gaps and contribute to previous studies, the key highlights of this research are.

1. Previous models and studies in Iran have often focused on a specific technology, such as electrolyzers, or have examined a particular region or area of Iran. In contrast, this study is a comprehensive national model that integrates all parts of the hydrogen supply chain with the energy supply system, enabling a thorough and complete technical and economic evaluation of hydrogen production in Iran.
2. The above study can provide a precise and quantitative assessment of the impact of hydrogen production on other energy carriers. For example, hydrogen production from natural gas can affect the production, export, and import of gas, electricity, etc. throughout the country's energy supply system.
3. This study aims to consider environmental parameters in addition to the technical and economic evaluation of scenarios and technologies and proposes policies to reduce emissions in the hydrogen supply chain and the overall energy supply system at the national level. Furthermore, by conducting sensitivity analyses on certain parameters, it seeks to identify the best scenario from both an economic and environmental perspective.

In the next section, the research methodology, including the structure and concepts of the modelling performed, along with the policies and hydrogen production chain, will be presented. Thereafter, the research findings will be examined, and finally, the conclusions and suggestions derived from this research for use in future studies will be explained.

2. Methodology

In the context of energy systems, energy models are used to evaluate and make decisions within the energy supply and demand system. OSeMOSYS (Open Source Energy Modeling System) is an open-source and freely accessible bottom-up modeling framework designed for the long-term optimization of the energy system and energy mix in user-defined areas [61]. Implementing various scenarios, calculating energy demand by sector and technology, minimizing system costs, integrating with other models, and quickly adapting its structure are some of the capabilities of this model [62]. Therefore, the OSeMOSYS model has been widely used by researchers and energy planners to identify optimized transition pathways in the energy supply sector, while also evaluating the environmental, economic, and social impacts of large-scale energy system adjustments [63].

OSeMOSYS optimizes the energy supply system by minimizing investment, fixed, and variable costs for each technology to meet energy demands. If carbon taxes or emission costs are considered in the model, they will also be included in the objective function. All costs are then discounted to the base year and subtracted from the salvage value. Further details about the model structure and formulation are available at [64].

2.1. Reference energy system

The reference energy system includes a variety of energy carriers that link many technologies at various levels, from energy resources to primary, secondary, and final energy (or useful energy). Iran is one of the most diverse countries in terms of energy carrier usage. The presence of a complete supply chain of fossil resources and an extensive, intricate renewable potential makes energy modeling more challenging. According to Fig. 2, energy carriers are depicted as vertical lines, while energy technologies are represented as horizontal lines. Any technology can receive either a single energy carrier or a set of energy carriers to produce either one or a set of energy carriers.

Energy technologies can be classified into four main categories. The first category, energy resource exploitation technology, encompasses extraction and usage, involving both renewable and non-renewable resources. This category includes hydropower, geothermal, wind, solar photovoltaic, and solar thermal power plants. It also encompasses the extraction of crude oil and natural gas, both offshore and onshore; the gathering and processing of biomass resources; the mining and enrichment of uranium; and the extraction and processing of coal. The second category of technologies consists of conversion processes designed to produce secondary energy carriers in the required forms. This includes nuclear power plants, waste incinerators, coke production facilities, gas injection, gas flaring, gas processing plants, LNG processing plants, oil, and condensate refineries, and various types of thermal power plants such as gas turbines, steam turbines, combined cycle plants, etc. In the third category, technologies for supplying energy carriers to final consumers are classified into transportation/transmission, import/export potential, and distribution. This encompasses coal, natural gas, oil, natural gas condensate, petroleum products, LNG (Liquified Natural Gas), and electricity technologies. Lastly, end-use technologies are utilized to deliver energy services; however, this research only considers energy demand up to the point of final energy consumption. It is worth noting that given the significant importance of the petrochemical industry in Iran, energy demands in this industry have been calculated separately and incorporated into the model.

In Fig. 2, the technologies used in the hydrogen supply chain are highlighted in red and are integrated with other energy technologies. According to this figure, hydrogen supply modeling has been conducted alongside the modeling of Iran's entire energy system to identify opportunities and potential for expanding the hydrogen industry within Iran's energy supply system, and to reveal the impacts of hydrogen production on the overall energy supply system under various scenarios.

2.2. Hydrogen supply chain

The hydrogen supply chain plays a vital role in the emerging clean energy transition, where hydrogen is gaining attention as a flexible and low-carbon energy carrier. In this context, understanding the various methods and technologies for hydrogen production is essential, as each

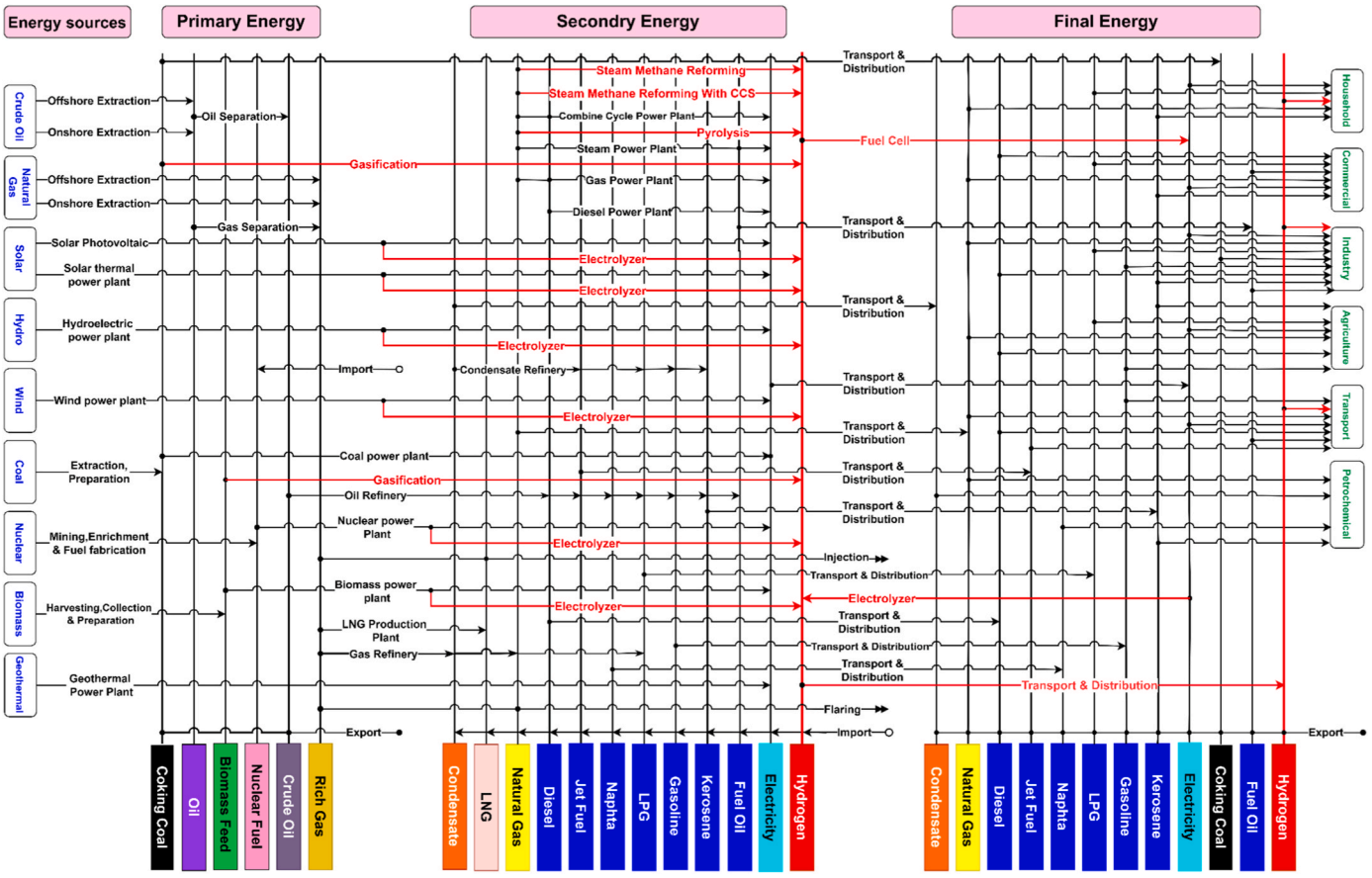


Fig. 2. The Reference Energy System of the developed model.

has different implications for efficiency, cost, and environmental impact. Fig. 3 provides a more detailed view of the hydrogen supply chain that has been developed within the national energy supply model in this research.

According to this figure, seven primary sources for hydrogen production are considered in the model. Among these, natural gas is the most widely used source for hydrogen production. Iran is the second largest holder of natural gas reserves in the world. Moreover, the existence of natural gas production and distribution infrastructure in the country, as well as gas exports, presents significant opportunities for utilizing this energy carrier [65]. Furthermore, biomass and coal can also be used for hydrogen production. Electricity generation technologies, such as renewable energy technologies, nuclear power plants, and solar systems [66], can produce the necessary electricity for electrolyzers to produce hydrogen. Additionally, hydrogen can be produced using grid electricity derived from fossil fuel technologies such as gas and combined cycle power plants.

To better understand hydrogen technologies, Fig. 4 demonstrates a color-coded technology that is used to describe different hydrogen production processes and their environmental impacts. Each color represents a category based on the primary energy source and carbon emissions involved in hydrogen production [67].

Table 3 provides a more detailed explanation of the most important hydrogen production technology used by the methods previously mentioned in Fig. 4. It is worth noting that the color designations in the hydrogen rainbow may vary in different studies, but in this research, an attempt has been made to use the nomenclature and color assignments based on the most common research consistent with Fig. 4.

After production, the hydrogen storage technologies and subsequent transportation for consumption in various sectors are used. Similar to production methods, hydrogen has various transportation methods. These methods are vital for the hydrogen economy and ensure that hydrogen produced from various sources is transported safely and efficiently to the required locations. According to Fig. 3, four technologies for transporting hydrogen to consumers are seen in the model. The first technology is pipeline transportation. Similar to natural gas pipelines, this technology is the most efficient for large volumes over short and medium distances [75]. Costs can also be reduced by mixing hydrogen with natural gas and injecting it into the existing gas network, thus reducing the need for new pipelines tailored to pure hydrogen conditions. However, due to limitations on the concentration of hydrogen to prevent damage to infrastructure, there are certain constraints [76]. The second technology is liquefaction. In this condition, hydrogen is transported in a liquid state at a temperature down to −253 °C. This method is energy-intensive and costly due to the liquefaction process, but it occupies less space for long-distance transportation. The next technology is the compression of hydrogen and its road transportation in high-pressure tanks [77]. The last technology is the conversion of hydrogen to ammonia and its transportation to the required centers. Since ammonia is much easier and faster to transport, this method can transport a large volume of the required hydrogen, but this method is energy-intensive, expensive, and has relatively high energy losses in the overall conversion process [78]. In addition to the aforementioned technologies, at the consumption side, fuel cell technology for generating electricity from hydrogen is also included in the model.

It should be noted that the process of hydrogen storage is at the beginning of the transportation process, and the technical and economic indicators of these technologies, along with transportation technologies, are considered in the developed model. Each of the hydrogen production and transportation methods has its advantages and disadvantages in terms of cost, energy efficiency, safety, and infrastructure requirements. To accurately determine the advantages and challenges of each, it is

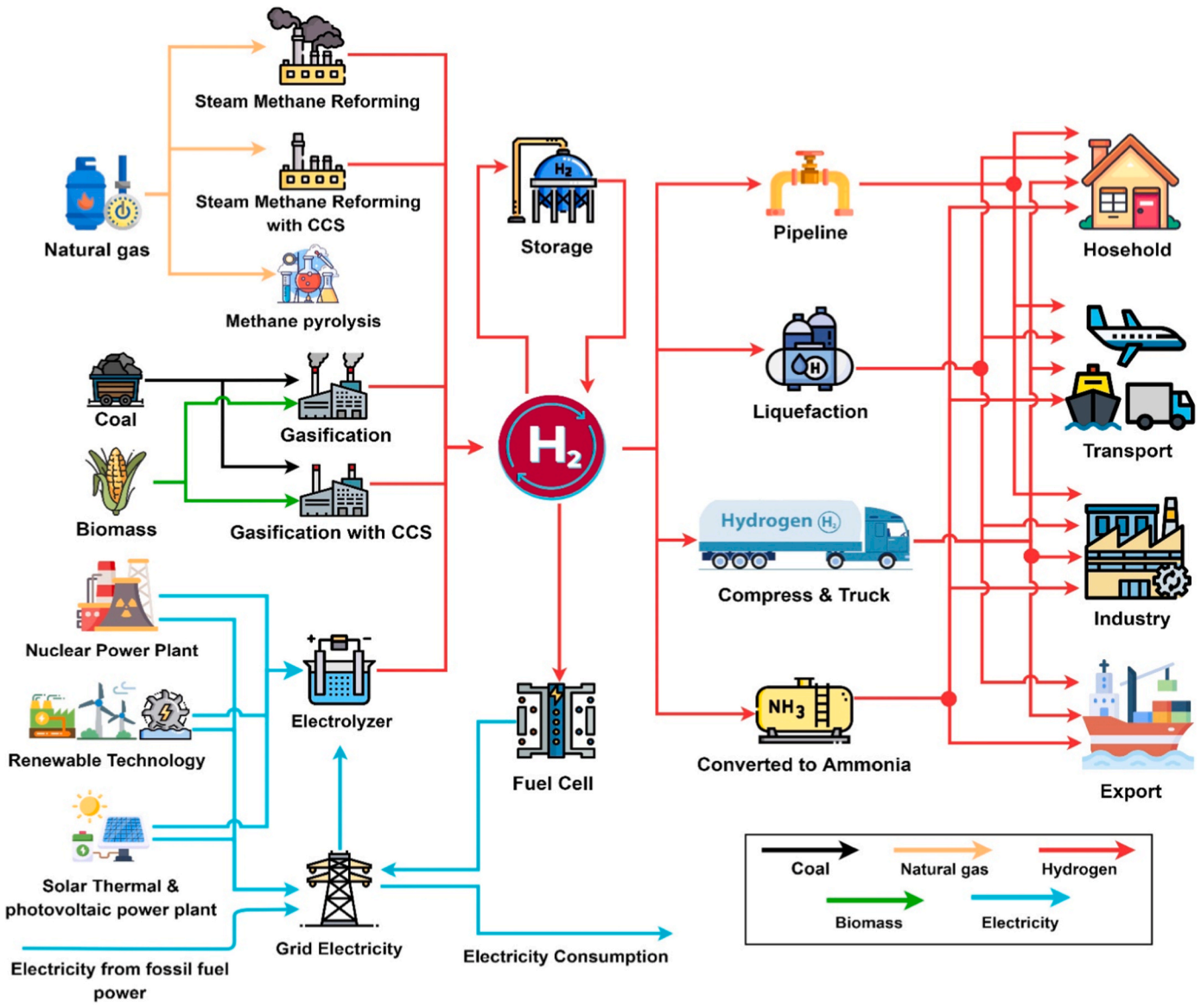


Fig. 3. Hydrogen supply chain in the developed model.

necessary to place all the above technologies alongside the entire energy supply system to determine the optimal hydrogen production methods and the impacts on other energy sectors using the energy systems modelling and optimization.

2.3. Demand projections

Iran's energy demand in this study comprises six sectors: household, industrial, transportation, agriculture, commercial, and petrochemical. The demand modeling is done using the Leap model [79]. Projections from the Fertility Bureau of the Statistical Center of Iran (SCI) are used to project demographic trends, rates of urbanization, and average household size that are consistent with an average GDP growth rate of 3.5% through 2050. Also, the Iranian Fuel Conservation Company (IFCO) provides statistics on energy intensities and fuel shares in each branch, derived from energy audit programs, for the transportation, petrochemical, and industrial sectors. Additionally, data on energy usage for the commercial and agricultural sectors is sourced from the 2021 energy balances [80]. Value-added figures, which include activity level statistics for all sectors except households, are obtained from the Central Bank of Iran's (CBI) publication of the national account balance for the year 2023 [81]. Table 4 shows the total demand for each energy carrier. Based on the results, demand for most energy carriers is increasing. In particular, natural gas has a very high growth rate, which shows its importance as an energy carrier in the future.

Specifically, hydrogen also has a significant demand in three sectors: household, industry, and transportation. Fig. 5 shows the hydrogen demand by different sectors. In the household sector, hydrogen can contribute to meeting the demand for heating, cooking, and water heating. In the industry, hydrogen can be used in oil refining, producing lighter hydrocarbons, manufacturing chemicals such as ammonia and methanol, and steel production. Additionally, the use of various hydrogen-powered vehicles can drive hydrogen demand in the transportation sector [82]. Furthermore, hydrogen can be used with fuel cell technology to meet a portion of the electricity demand in all energy-consuming sectors.

2.4. Assumptions and data

In order to achieve logical and accurate results, assumptions have been considered in the developed model, which will be explained below.

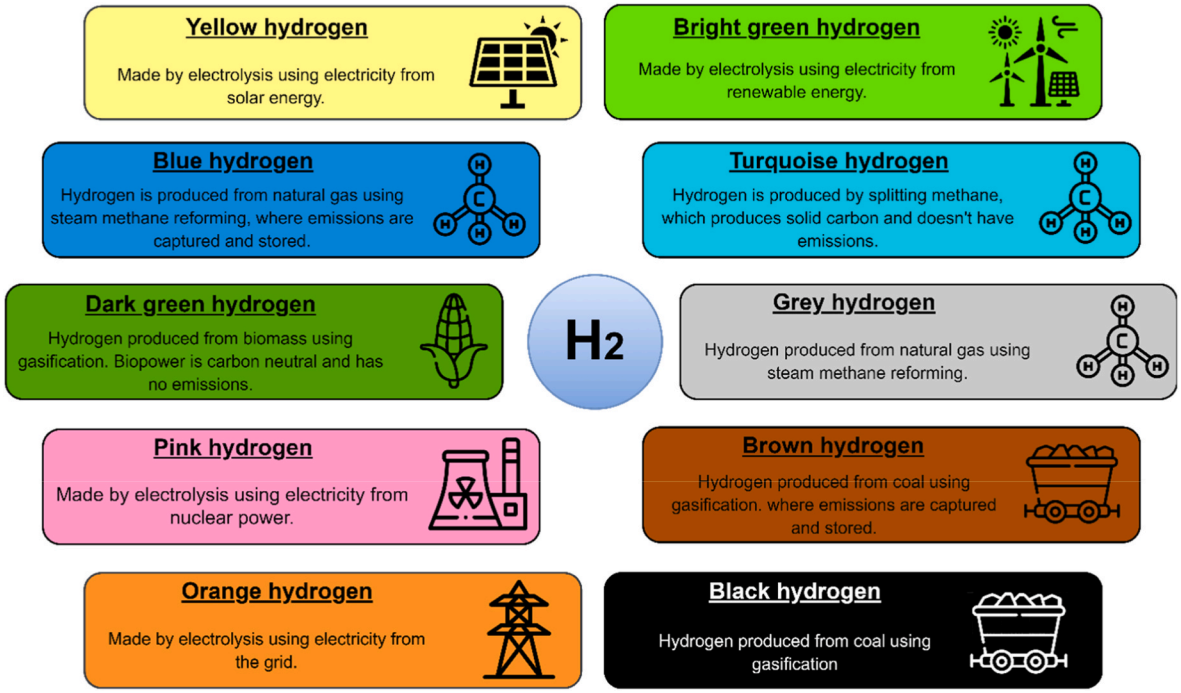


Fig. 4. Hydrogen production methods that are considered in the model.

1. The modeling currency is in US dollars ($), and an annual discount rate of 8% is assumed.
2. To verify the model's results against actual historical data for the first years of modeling, the modeling period has been selected from 2020. Furthermore, to evaluate and check the model's correct performance, a sensitivity analysis was conducted between the carbon tax rate and hydrogen production from various technology mixes in 2055. This was done to further examine the model's behavior. Fig. 9 shows the results of this sensitivity analysis.
3. The values of energy carrier exports and imports are endogenously calculated within the model. This means that the model allows for the import of all energy carriers if necessary, and it can also produce energy carriers in excess of demand and export the surplus if deemed appropriate. Since the objective function of this model is cost minimization, the model is first required to meet domestic energy demand and subsequently generates energy for export only to the extent that it remains economically viable. The inclusion of production costs ensures that energy exports do not increase infinitely, allowing the model to achieve a stable optimal solution for each year.
4. Due to natural gas supply imbalances in Iran, an alternative fuel is sometimes used for electricity generation in fossil fuel power plants. Therefore, two operating modes have been defined for them, and parameters such as emission factors, variable costs, efficiency, and others have been considered separately for each mode.
5. The model enables upgrading two technologies. Gas power plants have the potential to be converted into combined cycle power plants. The model can, at its discretion, construct a technology to capture and recover the waste heat from gas power plants, thereby generating additional electricity. The model can also utilize carbon dioxide capture and storage technology alongside steam methane reforming technology to absorb and store the emissions from this process, effectively transforming grey hydrogen into blue hydrogen. It is worth mentioning that the two technologies of combined cycle power plants (CC PP) and steam methane reforming (SMR) with carbon capture and storage (CCS) have been considered independently in the model, as well.
6. As shown in Table 5, the parameter 'Year Split' indicates the time division of each modeled year, referring to the model's time resolution. Given the high importance of electricity and gas production during peak consumption periods, the specified profile demand for those in the model must have high accuracy. Therefore, considering the electricity and gas consumption statistics in Iran and the country's climatic conditions [83], each year is divided into two seasons (hot and cold), and each day is divided into four parts. It's worth noting that the specified profile demand of other carriers is also included in the model.

Given that the developed model is a bottom-up model, it consists of a vast amount of data and parameters. Key technical and economic data of the technologies included in this model are provided in a supplementary data file available at [84]. These data were collected from international sources and annual reports of the Iranian Ministry of Petroleum and the Ministry of Energy [80] and then, validated. Considering that many technical and economic parameters of technologies, especially renewable technologies (such as efficiency, capital costs, etc.), will change according to the forecasts of the IEA [85] and IRENA [86], these changes have been considered in the model from 2020 to 2055.

2.5. Scenarios

To better evaluate the pathways and methods of hydrogen production in Iran and the impacts this carrier has on Iran's overall energy supply system, the developed model has been solved and analyzed under five scenarios. Additionally, some of these scenarios are further divided into sub-scenarios, which will be explained below. It should be noted that the equations specified in each scenario are existing components of the OSeMOSYS framework, which have been adapted to align with the low-carbon scenarios defined for this study.

- **Business as usual scenario:** In this scenario, energy system optimization is conducted based on current conditions, and no limitations are imposed on any technologies or energy carriers. The linear optimization will solely focus on minimizing costs, as defined by the model's objective function.
- **Carbon tax scenario:** A tax on greenhouse gas emissions is an effective tool in environmental policy. By placing a value on

Table 3
Technologies that are considered for hydrogen production in the model.

Hydrogen Type	Production
Green Hydrogen (Dark & Bright)	The cleanest type of hydrogen is produced by electrolyzing water according to the following equation, using electricity generated from renewable sources such as wind, hydroelectric, or geothermal power, and producing no pollutants [68]. $2H_2O_{(L)} \rightarrow 2H_{2(g)} + O_{2(g)}$ Additionally, since the process of producing hydrogen from gasification of biomass is carbon neutral, the hydrogen produced from this process is considered green hydrogen. However, according to Fig. 2, the biomass gasification process has been considered in the model both conventionally and with carbon capture and storage technology [69].
Grey Hydrogen	Most of the hydrogen produced today is made from natural gas through a process called steam methane reforming, which is represented by the following equation. However, this process releases carbon dioxide into the atmosphere [70]. $CH_{4(g)} + 2H_2O_{(g)} \leftrightarrow CO_{2(g)} + 4H_{2(g)}$
Blue Hydrogen	Like grey hydrogen, it is produced from natural gas, but by employing carbon capture and storage technologies, less carbon dioxide is released [71].
Turquoise Hydrogen	Using pyrolysis technology, natural gas is split into hydrogen and solid carbon. The significant advantage of this method is that there is no CO_2 emission and the produced carbon is in solid form [72]. $CH_{4(g)} \rightarrow C_{(S)} + 2H_{2(g)}$
Yellow & Orange Hydrogen	Typically refers to hydrogen produced through electrolyzers using solar power. However, sometimes, in addition to solar electricity, grid electricity is also used to produce yellow hydrogen. It's worth noting that although the hydrogen produced from solar power and grid electricity is named the same, their production sources are considered separately in the model [73]. To provide better clarity, hydrogen produced using electricity from the grid is referred to as 'Orange hydrogen'.
Pink Hydrogen	This type of hydrogen is produced through electrolysis using electricity generated from nuclear power. Although this process does not produce CO_2, the use of nuclear energy raises concerns about environmental and safety issues [7].
Black & Brown Hydrogen	This hydrogen is produced from coal gasification, a process associated with high CO_2 emissions. Although there is no specific name for hydrogen produced from coal gasification with carbon capture and storage technology, this technology has been considered in the developed model [74], and in this case, the hydrogen produced is called brown hydrogen in the model.

greenhouse gas emission tax, this creates a financial incentive to reduce carbon footprints. This means that products and services with higher emissions will generally have a higher cost, making their true impact on the environment visible. In recent years, the carbon tax rate has varied considerably across countries, with European Union leading the way in implementing this policy [87]. Given that a carbon tax does not currently exist in Iran, the implementation of these rates has been modeled to begin in 2025 and increase gradually. Accordingly, four rates of 10, 30, 60, and 100 USD/tCo_2 equivalent were applied as carbon price in the model, and their results were evaluated. According to Equation (1), the greenhouse gas emissions, considering their price as a carbon tax, determine the technology emissions penalty. This amount will be added to the objective function along with other costs.

$$\forall_{y,t,r} \quad Minimize \sum_{y,t,r} TotalDiscountedCost_{y,t,r} = DiscountedOperatingCost_{y,t,r} + DiscountedCapitalInvestment_{y,t,r} - DiscountedSalvageValue_{y,t,r} + \boldsymbol{DiscountedTechnologyEmissionsPenalty_{y,t,r}} \tag{1}$$

❖ **Low-rate facilitation scenario:** Facilitations for the hydrogen industry have been implemented in the form of reducing the capital cost of hydrogen technologies. Equation (2) shows the Capital Recovery Factor (CRF), where i represents the discount rate and n is the system's lifetime. This parameter is a factor that, when multiplied by the initial capital cost, determines the annual payback amount and, in a way, indicates the initial investment share of the project's overall financial balance [88].

$$CRF = \frac{i(1+i)^n}{(1-i)^n - 1} \tag{2}$$

To apply facilities to hydrogen industries, a minimum discount rate (0.25%) has been considered, and in this case, there will practically be no inflation when calculating costs. Therefore, the amount of capital costs will be evenly distributed without a decrease in value in the coming years. Considering the new discount rate, the new CRF has been obtained, and according to Equation (3), the new cost of investing in hydrogen technologies will be calculated and incorporated into the model.

$$\frac{CRF_2(with\ r = 0.25\%)}{CRF_1(with\ r = 5\%)} \times Capital\ Cost_1 = Capital\ Cost_2 \tag{3}$$

To conduct a more thorough examination of the provision of facilities to hydrogen fuel production technologies, the aforementioned scenario has been applied once exclusively to hydrogen production technologies (HPF) and once to the entire hydrogen supply chain (HVCF). In other words, in the first case, the mentioned facilities are only provided to technologies such as steam methane reforming, electrolysis, pyrolysis, and so on. However, in the second case, for example for the production of green hydrogen, the facilities, in addition to electrolyzers, will also be allocated to renewable technologies for generating electricity used for green hydrogen production.

❖ **Emissions cap scenario:** Iran in the latest Intended Nationally Determined Contribution (INDC) published in 2015, committed to

Table 4
The final energy demand.

Energy demand (PJ/y)	2020	2025	2030	2035	2040	2045	2050	2055
Condensate	511.56	656.1	844.28	1083.49	1380.74	1743.12	2177.73	2691.65
Diesel	295.57	316.03	337.10	359.39	383.19	391.29	418.91	448.30
Fuel Oil	236.33	235.33	235.27	235.62	236.14	236.74	237.33	237.90
Gasoline	1143.76	1381.5	1588.41	1764.49	1909.77	2024.24	2107.91	2160.77
Jet Fuel	61.76	67.40	72.89	78.24	83.45	88.52	93.44	98.23
Kerosene	152.03	168.96	186.81	205.58	225.27	245.89	267.40	289.84
Naphtha	71.17	89.66	115.71	149.33	190.52	239.28	295.61	359.51
LPG	30.87	33.61	36.77	40.36	44.36	48.79	53.64	58.92
Coal	96.18	105.32	114.10	125.68	137.85	151.98	168.53	187.97
Natural gas	4661.74	5196.34	5702.83	6202.04	6662.74	7013.90	7552.08	8025.35
Electricity	936.23	1018.63	1103.81	1192.68	1285.76	1383.40	1485.87	1593.43
Hydrogen	4.94	55.40	140.23	297.41	482.73	636.06	873.92	1159.85

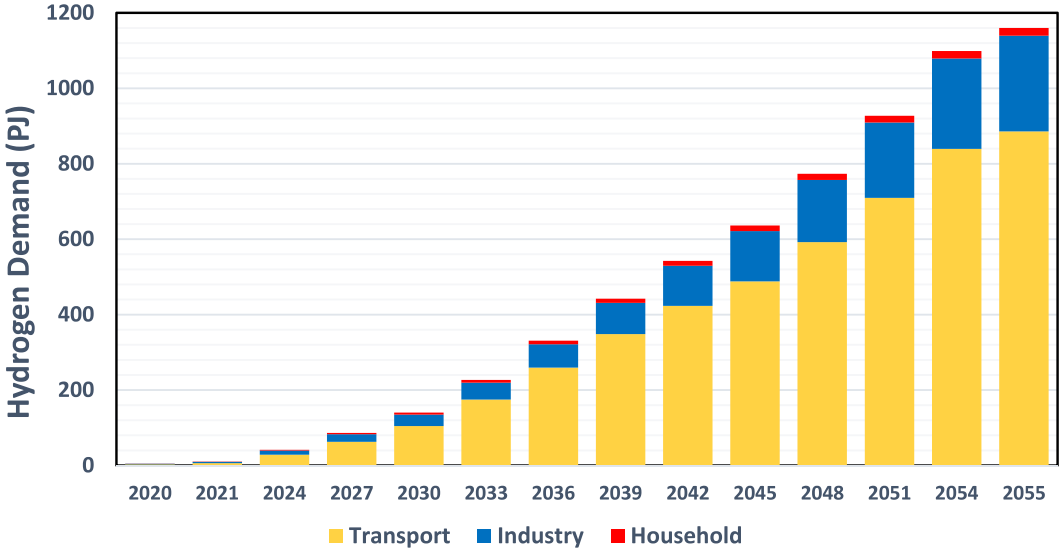


Fig. 5. Hydrogen demand in sectors.

Table 5
Year Split values in the model.

No	*Time Slices*	2020	2022	…	2054	2055
1	Hot-day-Intermediate	0.208	0.208	0.208	0.208	0.208
2	Hot-day-Peak	0.039	0.039	0.039	0.039	0.039
3	Hot-night-Intermediate	0.208	0.208	0.208	0.208	0.208
4	Hot-night-Peak	0.045	0.045	0.045	0.045	0.045
5	Cold-day-Intermediate	0.158	0.158	0.158	0.158	0.158
6	Cold-day-Peak	0.050	0.050	0.050	0.050	0.050
7	Cold-night-Intermediate	0.220	0.220	0.220	0.220	0.220
8	Cold-night-Peak	0.072	0.071	0.071	0.071	0.071

reducing its emissions by 4% below business-as-usual by 2030, and by 12% if financing is provided by international institutions and organizations [14]. Assuming that this goal is adopted in the country starting today, the country's emissions should be reduced by 40% compared to business-as-usual in 2050. Therefore, the constraint on reducing CO_2 emissions in this scenario is a 40% reduction in 2050 compared to business as usual, and this constraint has been applied to the model from 2025 onward. According to Equation (4), a constraint on annual emission production will be activated in this scenario, which will not allow the amount of emissions to exceed the permitted limit each year.

$$\forall_{y,e,r} \quad AnnualEmissions_{y,e,r} \leq AnnualEmissionLimit_{y,e,r} \tag{4}$$

- **Energy desubsidization scenario:** Fossil fuel subsidies, by lowering their prices, increase the consumption of energy carriers and have detrimental effects on the environment. Iran, after Russia, pays the highest energy subsidies in the world [89]. This has caused the domestic prices of energy carriers to deviate significantly from global prices, leading to excessive domestic consumption, energy inefficiency, and environmental concerns. In this scenario, as recommended by the IEA, the energy carriers in the country are gradually desubsidized, with prices aligned increasingly closer to global market levels.

The key points of the scenarios are summarized in Table 6.

3. Results and analysis

This section first examines the results of Iran's whole energy supply system in the business as usual (BAU) scenario and then compares hydrogen production and various technical and economic indicators across different scenarios.

Table 6
Names and important points of scenarios.

Scenario name	Abbreviation	Key Assumptions
Business as usual scenario	BAU	Optimization of the energy supply system based on current conditions
Carbon tax scenario	CTS10	Carbon price of 10 USD/t applied in the model
	CTS30	Carbon price of 30 USD/t applied in the model
	CTS60	Carbon price of 60 USD/t applied in the model
	CTS100	Carbon price of 100 USD/t applied in the model
Low-rate facilitation scenario	HPF	Provision of financial incentives to technologies for hydrogen production and fuel cell development (Hydrogen production finance)
	HVCF	Provision of financial incentives to the entire value chain of hydrogen production, distribution, and utilization (Hydrogen value chain financing)
Emissions cap scenario	ECS40	A mandated 40% reduction in CO_2 emissions relative to a BAU scenario
Energy desubsidization scenario	EDS	Removal of subsidies on fossil fuel carriers and aligning their prices with global market projections

3.1. National energy supply system

Fig. 6 shows the final production of all energy carriers from 2020 to 2055 in the BAU scenarios. According to this figure, the production of almost all energy carriers, especially petroleum products, is increasing. According to the forecasts, if Iran's capacity for natural gas production is not maintained, the country will face a sharp decline in natural gas production in the coming decades. This is also evident in Fig. 6, where natural gas production will peak around 2029 and decline thereafter. The results indicate that while electricity production is projected to increase in the coming years, the model's preference for combined cycle power plants and the anticipated decline in natural gas production will cause the growth rate of electricity to slow down after 2033. Therefore, from 2047 onwards, this results in significant growth in coal-fired power

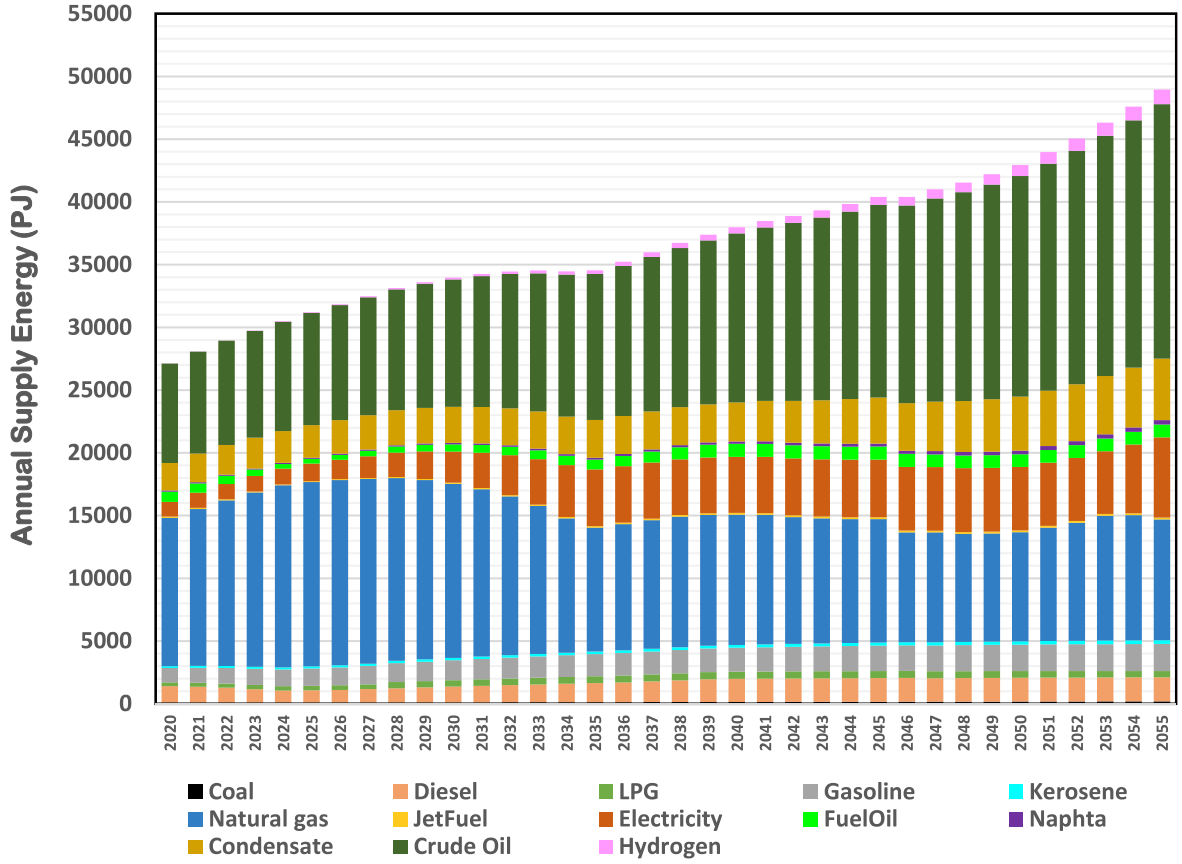


Fig. 6. Final production of all energy carriers in the BAU scenario.

plants and renewable technologies (Fig. 7). It should be noted that Fig. 6 only shows the production of energy carriers in the final energy sector and it does not reflect the demand of any one energy carrier. For example, the figure given for crude oil is only for the final energy sector (which consists of crude oil exports and non-energy uses), and crude oil required as feedstock for refineries are not included in it.

Fig. 7 presents the electricity generation from each technology over the modeled years. According to this figure, in the first decade of the modeling period (2020-2030), the model optimally decides to avoid the installation of OCGT (Open Cycle Gas Turbine) and convert its residual

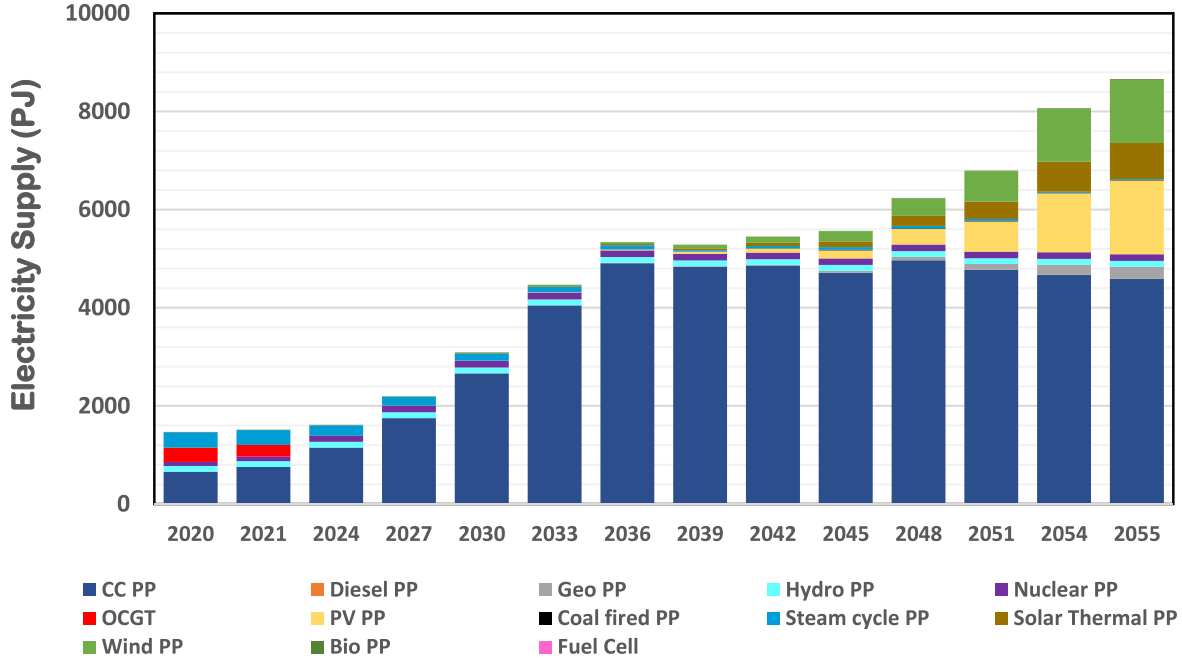


Fig. 7. Electricity supply by source in the BAU scenario.

capacity to CC PP. Additionally, the capacity of nuclear and hydroelectric power plants is also increasing. Subsequently, given the high proportion of electricity generation supplied by CC PP, the decrease in natural gas production in the second decade (2030-2040) of the modeling period will slightly reduce the growth rate of electricity generation compared to previous years. However, in the last decade of the modeling period (2040-2055), with the increase in the capacity of renewable systems, electricity generation will increase significantly. Therefore, in 2055, photovoltaic, solar thermal power plant, and wind power plant will account for 17.37%, 8.4%, and 14.84% of the total electricity generation, respectively. It is obvious that with the implementation of low-carbon policies (such as carbon taxes), electricity generation from the combined cycle will decrease and electricity generation from renewable technologies will increase. The results also show that the model generally does not select fuel cell technology to meet electricity demand. This is predictable in the BAU scenario, as given the domestic natural gas price, the economic efficiency of converting gas to electricity is much higher than converting gas to hydrogen and then hydrogen to off-grid electricity. Moreover, the high physical capacities of renewable technologies in Iran (especially solar technologies), the decrease in capital costs, and the improvement of technical efficiencies of renewable sources will facilitate their development in the future.

From the environmental point of view, Fig. 8 presents CO_2 emissions from each technology in the BAU scenario. According to this figure, the majority of emissions originate from electricity generation, specifically from combined cycle technology. Given that the growth of combined cycle technology stagnates in the final years of the modeling period, emissions will follow an almost constant trend.

3.2. Hydrogen intrusion in the national energy supply system

Fig. 10 illustrates the hydrogen production from each source in every scenario. The results indicate that under the BAU scenario, all hydrogen demand will be met by steam methane reforming (grey hydrogen). This is expected given the very low price of gas in Iran compared to other countries and the lack of environmental restrictions on emissions. Even today, despite the rapid growth of renewable technologies, a large portion of the world's hydrogen demand is met through natural gas and steam methane reforming [90]. However, in the final period (2040-2055), due to the decreasing costs of renewable technologies and the increasing natural gas imbalance in the country, a portion of hydrogen production will come from electrolyzers powered by solar energy, such that in 2055, 4.32% of the total production will be yellow hydrogen. Also, in this scenario, hydrogen export is increasing and will reach 385 PJ in 2055.

Subsequently, following the carbon tax imposition, the model prioritizes hydrogen production using lower-emission technologies. The higher the carbon tax, the greater the incentive for the model to increase the capacity of new technologies and reduce emissions. Results indicate that when the carbon tax is 10 $/t to 30 $/t, the model attempts to convert a portion of hydrogen production from grey to blue by constructing CCS technology, thereby reducing emissions. However, when the tax reaches 60 $/t to 100 $/t, the model will proceed to construct pyrolysis technology, which has no emissions. In scenario CTS100, all production will be turquoise hydrogen from 2049 onwards. It should also be noted that due to the substitution of new and costly technologies due to the imposition of a tax, hydrogen production and consequently its exports will decrease to higher carbon prices. For instance, in scenarios CTS10, CTS30, CTS60, and CTS100, hydrogen export in 2055 will reach 225, 149, 96, and 87 PJ, respectively.

Given the significance of carbon taxation and the fact that this policy has been increasingly prioritized in the agendas of numerous countries in recent years, a sensitivity analysis between hydrogen production (including the technology mix) and the carbon tax rate was conducted. According to Fig. 9, when a 10 USD/tCO_2 carbon tax is applied, the model tries to reduce some of its emissions by converting grey hydrogen to blue hydrogen. To achieve this, it builds CCS technology, but it still uses SMR for hydrogen production.

As the tax rate increases, the model shifts toward using lower-emission technologies to meet hydrogen demand. When the tax reaches 40 USD/tCO_2, the model switches to pyrolysis and methane gas separation, which has zero emissions. Despite this method being pricier, the model chooses turquoise hydrogen over grey hydrogen to reduce the emissions penalty. As the tax rate goes up further, blue hydrogen production also decreases. The cycle continues until the model, with a 100 USD/tCO_2 tax, chooses pyrolysis to meet all its hydrogen demand in 2055.

Furthermore, in the HPF scenario, where low-rate loan incentives are provided only to hydrogen technologies, hydrogen production through coal gasification increases significantly. The capacity of this technology rose from 0.01 GW in 2020 to 9.36 GW in 2055, accounting for 83.11% of total hydrogen production in that year. This leads to a significant

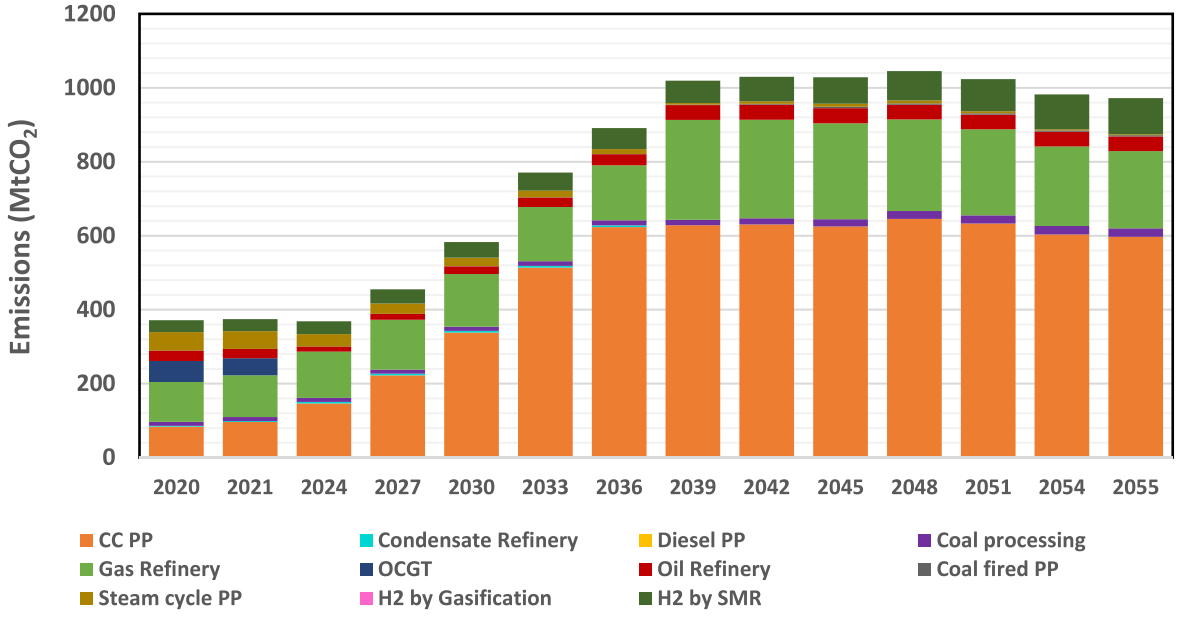


Fig. 8. Emissions by source in the BAU scenario.

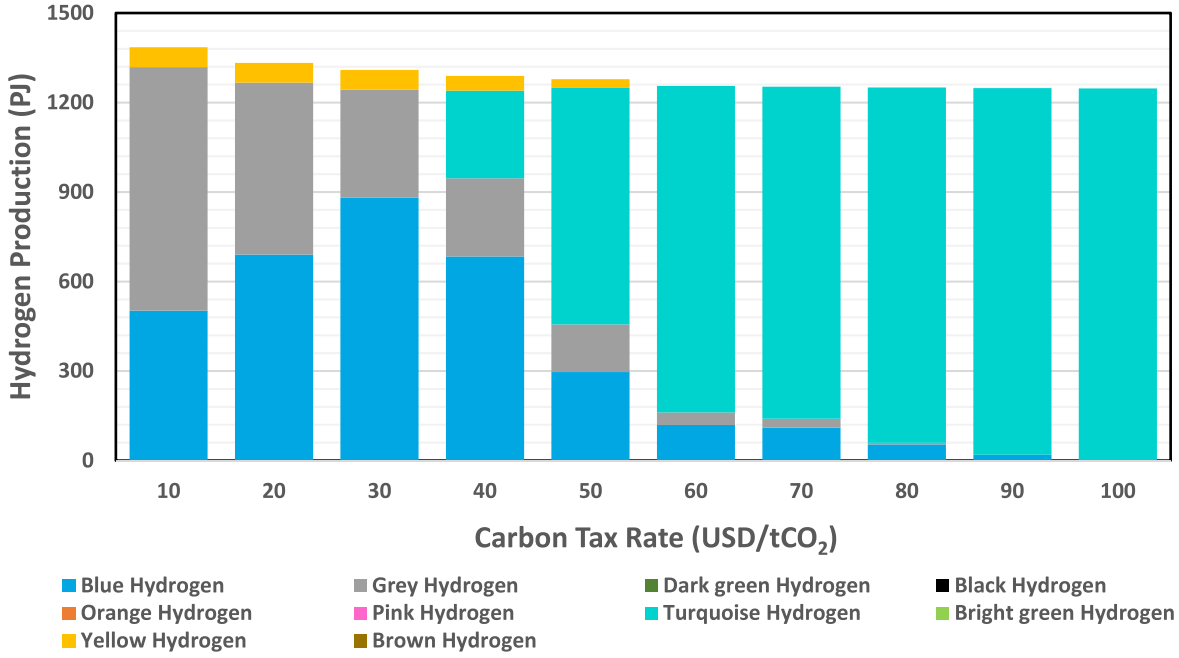


Fig. 9. Sensitivity analysis between hydrogen production and carbon tax rate in 2055.

increase in both hydrogen production and exports, with hydrogen exports reaching 441 PJ in the final year (2055). According to Fig. 10, the production of yellow hydrogen in this scenario has also increased by 22.39% compared to the BAU scenario.

However, when financial incentives are provided to the entire hydrogen supply chain (HVCF), electrolyzer technology coupled with renewable and solar electricity will experience significant growth. In 2055, these two technologies will account for 42.81% and 32.24% of total hydrogen production, respectively. In this scenario, we will witness the highest levels of hydrogen production and exports, with total annual hydrogen production reaching 2240 PJ in the final year of the modeling period, of which 1080 PJ will be exported. Additionally, black hydrogen production will also increase compared to the BAU scenario.

In the cap on total emissions scenario, the model decides to continue using natural gas as the primary energy source for hydrogen production. However, given the mandatory 40% reduction in emissions compared to the BAU scenario by 2055, the model strives to utilize cleaner technologies such as pyrolysis and SMR + CCS. Initially, by installing CCS technology, the model converts 59.02% of grey hydrogen into blue hydrogen. However, due to increasing demand and the need to reduce emissions in the final years, the model increases the production capacity of turquoise hydrogen. By 2055, this technology will account for 70.13% of total production. Hydrogen exports in this scenario also decreased by 77.01% compared to the BAU scenario.

In the EDS scenario, the technology portfolio selected for hydrogen production is identical to the BAU scenario. Only the total hydrogen production decreases by 19.14% compared to the BAU scenario, reaching 1249 PJ in 2055. The results in this scenario indicate that when the price of fossil fuels increases, the price of electricity also increases, given that a significant portion of the country's electricity is generated from fossil fuel power plants. This, in turn, affects the entire energy supply system. Therefore, there is no substantial change in the energy supply portfolio; only the overall cost of the supply system increases. On the other hand, although many countries do not allocate subsidies to energy carriers, a large portion of the world's hydrogen production is derived from fossil fuels, especially natural gas. In this case, renewable and low-carbon technologies have small shares of this market. This confirms the accuracy of the results obtained. Moreover, this indicates that changes in the price of energy carriers have a greater impact on energy demand than on energy supply.

As mentioned earlier, this model considers four hydrogen transportation technologies. Results show that from 2020 to 2047, the model only selects compression in tanks followed by road transport. However, from 2048 onwards, a small amount of capacity for transportation via pipelines, liquefaction, and conversion to ammonia also grows, with each of these technologies accounting for 7.32%, 2.11%, and 1.06% of hydrogen transportation in 2055, respectively. It was also mentioned that before transporting hydrogen, hydrogen must be stored. Therefore, the technical and economic parameters related to hydrogen storage (such as capital costs, efficiency, etc.) in each of the technologies considered in the model for hydrogen transportation are taken into account.

3.3. Evaluation and comparison

In this section, the environmental and economic outcomes of the aforementioned scenarios are compared. The environmental indicator here is considered to be the total emissions of the entire energy supply system and the hydrogen supply chain. The economic indicator for comparing different scenarios will be the total cost of the entire energy supply system and the hydrogen supply chain. It should be reiterated that the optimization in the above model, whether in the BAU scenario or other low-carbon scenarios, is solely aimed at minimizing the total system costs rather than reducing emissions.

Fig. 11 illustrates the cumulative emissions of the total energy supply system and the hydrogen supply chain. According to this figure, the cumulative emissions of the energy desubsidization and BAU scenarios are very close, with the EDS scenario having only 1.73% lower emissions compared to the BAU scenario. On the other hand, the results show that the best scenario for reducing system supply emissions is the implementation of a carbon tax. Obviously, the higher the tax rate, the more the system will strive to reduce carbon dioxide emissions by utilizing low-carbon technologies and increasing their capacity. It can also be concluded that imposing a 40% reduction in emissions has a greater impact than imposing a tax of 10 $/t or 30 $/t, but imposing taxes of 60 $/t and 100 $/t would be more successful in reducing emissions. Furthermore, due to the provision of incentives, the use of fossil fuels, especially coal, has increased, leading to higher emissions in the HPF

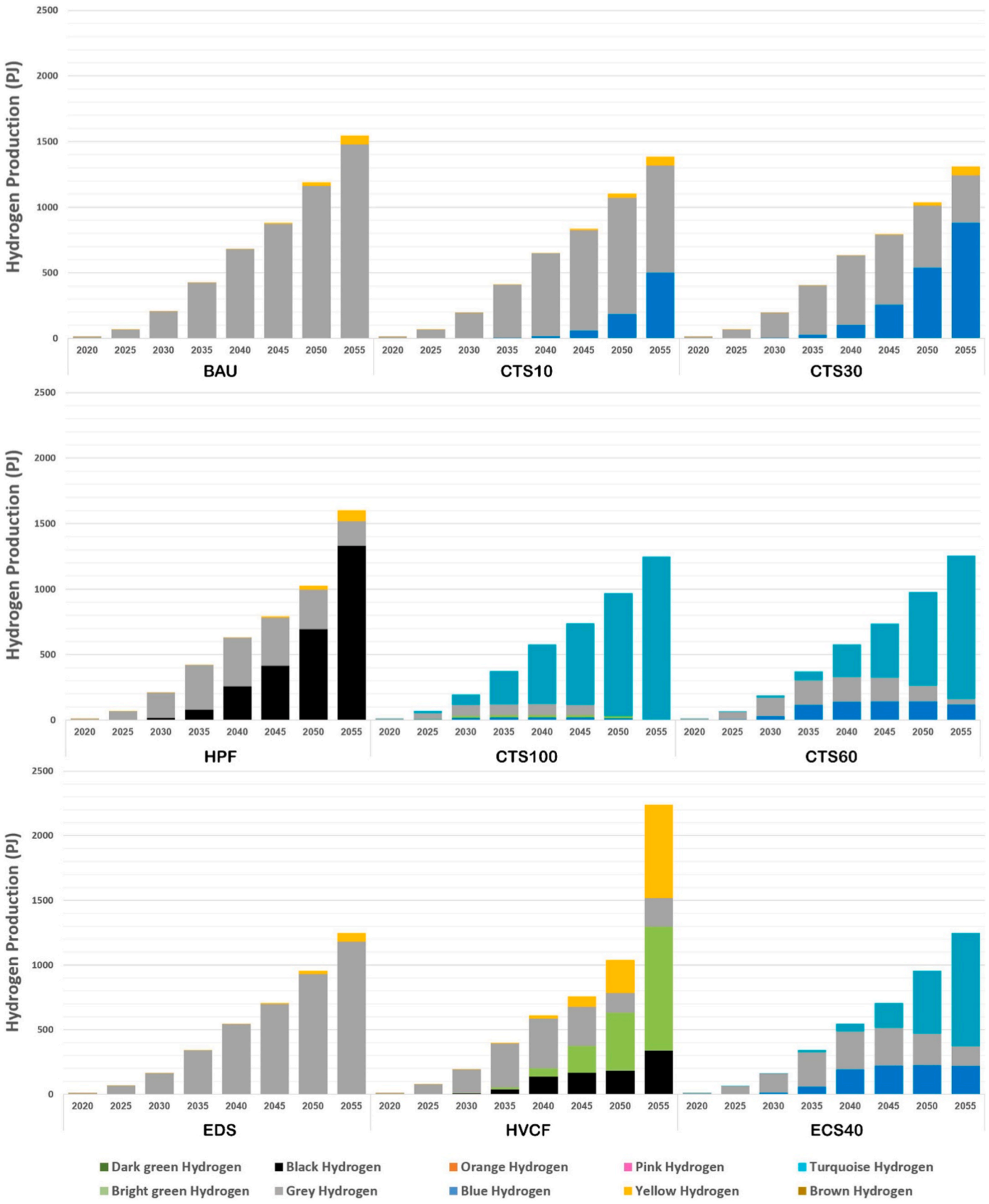


Fig. 10. Hydrogen production by source in modeled scenarios as defined in Table 6.

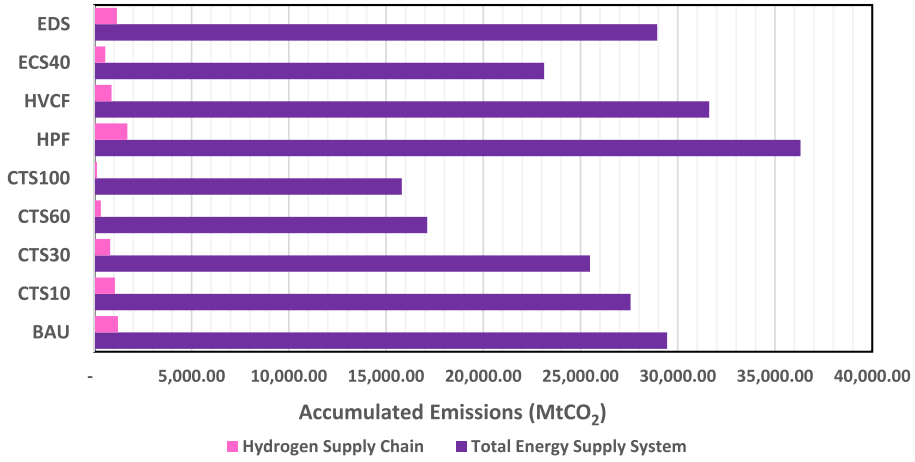


Fig. 11. Accumulated emissions from 2020 to 2055.

and HVCF scenarios compared to the BAU scenarios. Finally, it can be said that, in terms of emissions, the CTS100 scenario with a 46.31% reduction in the total energy supply system and a 88.78% reduction in the hydrogen supply chain compared to the BAU scenario is the best scenario, while the HPF scenario with a 23.31% increase in the total energy supply system and a 40.19% increase in the hydrogen supply chain is the worst scenario for this indicator.

To compare the costs in each scenario, the costs of the total energy supply system and the hydrogen supply chain are separated. Fig. 12 shows the discounted total costs of the total energy supply system. According to this figure, scenarios cap on emissions and CTS60 have the lowest total costs, resulting in cost reductions of 30.41 and 30.33 billion dollars, respectively, compared to the BAU scenario. It can also be said that in terms of cost, these two scenarios are very close. However, considering that 6.60% of the costs in the CTS60 scenario are made up of carbon taxes, if these amounts are used to compensate for the negative environmental impacts of the energy sector, it can be said that this scenario might be more impactful than the cap on emissions scenario. On the other hand, the EDS scenario has the highest cost compared to the BAU scenario, and as expected, given the removal of large energy subsidies in Iran in this scenario, the majority of costs in the scenario are variable costs. Another noteworthy point is that with the imposition of a carbon tax, the model at lower rates (10 $/t and 30$/t) has tried to meet energy demand without significant changes, especially in the energy supply system, and only reducing exports and installing CCS technologies are taken as priority measures. However, with a significant increase in the carbon tax rate, the model decides to increase the capacity of carbon-free technologies such as renewables or pyrolysis as quickly as possible and completely replace the previous technologies. Of course, in these scenarios, we will still see a decrease in energy exports.

Given that energy exports account for a large part of Iran's annual income, export revenues are an important metric for comparing different scenarios. Fig. 13 shows the discounted export revenues for the total energy supply system. According to this figure, most implemented scenarios exhibit a decrease in exports. This indicates that one of the reasons for the reduced costs in scenarios involving carbon taxes and emissions cap is the decrease in production and, consequently, energy exports. Only in scenarios HVCF and HPF an increase in energy exports of 9.97 and 8.85 billion dollars, respectively, compared to the BAU scenario is seen. Since only the hydrogen supply chain differs between these scenarios, the increased exports can be attributed primarily to the growth in facilitated hydrogen exports.

Fig. 14 illustrates the costs in each scenario within the hydrogen supply chain. According to this figure, the cost of hydrogen production has increased in all scenarios except HPF and HVCF compared to the BAU scenario, with capital costs constituting the largest portion of expenses. The hydrogen supply system cost has decreased by 0.06 and 0.24 billion dollars, respectively, in these two scenarios compared to the BAU scenario. Additionally, the variable costs of the HPF scenario are 1.59 times higher than those of the HVCF scenario due to the use of coal. On the other hand, results show that although the carbon tax rate in the CTS100 scenario is higher than in the CTS60 scenario, lower costs were incurred for it. This is because the model decided to initially replace the SMR technology with pyrolysis technology, which has no emissions,

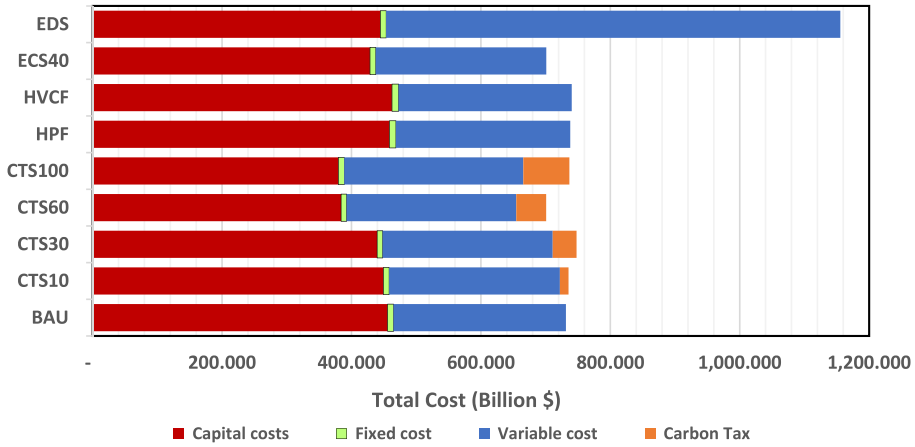


Fig. 12. Total supply energy system discounted cost from 2020 to 2055.

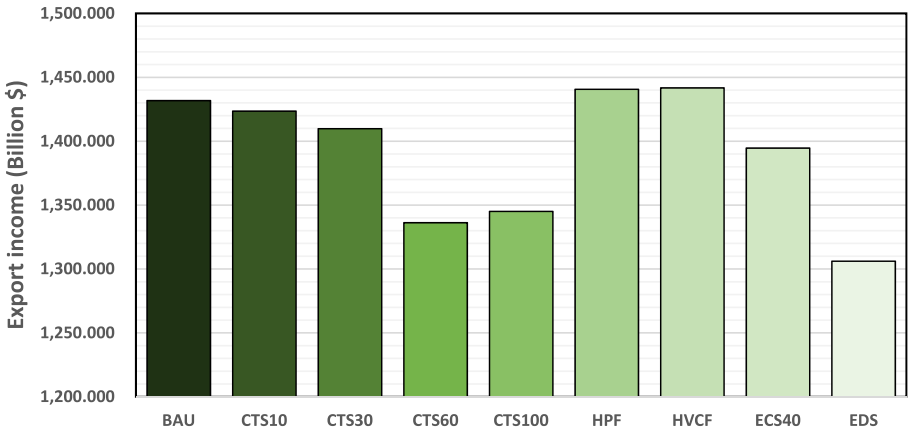


Fig. 13. Cumulative export income in the total supply energy system from 2020 to 2055.

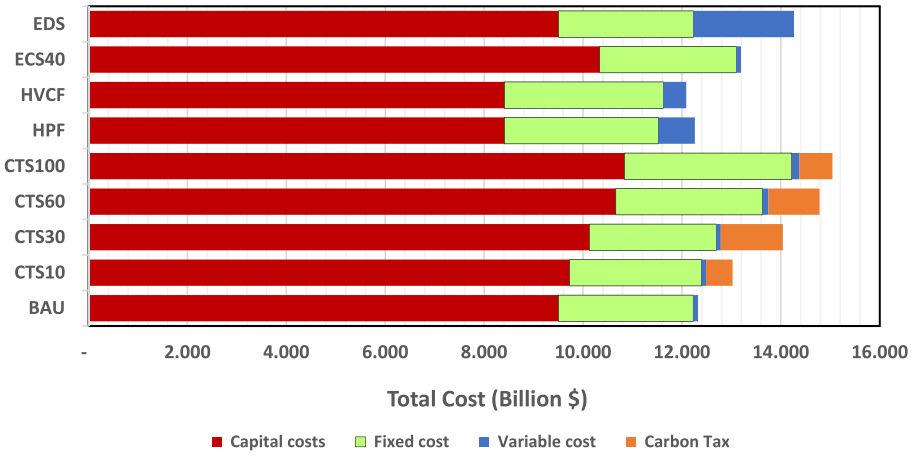


Fig. 14. Hydrogen supply chain cost from 2020 to 2055.

despite higher initial capital costs.

4. Discussions

In this section, the results of the scenarios are compared based on three indicators: GHG emissions, total costs, and export income, both for the total energy supply system and the hydrogen supply chain. Table 6 shows the performance of each scenario relative to the BAU scenario. If the scenario's performance has improved, it is shown in green, and if it has worsened, it is shown in red. The calculated values are in percentages, and the comparison benchmark is the BAU scenario. In all scenarios, the same level of final energy demand is exogenously given to the model. The further increase in export revenues also counts as an increase in energy production for export and earning income.

According to the results, greenhouse gas emissions have decreased in almost all scenarios, both in the overall energy supply system and in the hydrogen supply network. Only in the HPF and HVCF scenarios, greenhouse gas emissions increased compared to the BAU scenario. Since the HPF and HVCF scenarios only affect the hydrogen supply system, it can be said that by reducing the capital cost of hydrogen technologies, the model has proceeded to produce and consequently export more hydrogen, and this has caused the emissions from SMR technology to increase. However, specifically in the hydrogen supply chain, and because hydrogen production from coal gasification has grown by 83.11% in the HPF scenario, emissions have increased in this scenario. But in the HVCF scenario, since the model has decided to move towards renewable technologies from 2030, emissions have decreased by 27.28% compared to the BAU scenario. It's also clear that as the carbon tax rate increases, GHG emissions will decrease. However, this relationship is not linear.

According to Table 7, the percentage reduction in emissions from a carbon tax rate of 30 $/t to 60 $/t is 28.43% in the overall energy supply system and 39.37% in the hydrogen supply chain. However, the percentage reduction in greenhouse gas emissions from a rate of 60 $/t to 100 $/t, despite the larger increase in the tax rate, is only 4.43% and 16.30%, respectively.

Costs of the system also increase in almost all scenarios compared to the BAU scenario. Only in the CTS60 and ECS40 scenarios, due to a decrease in energy production and consequently a decrease in energy exports, system costs will decrease slightly. For example, it can be seen that although the tax rate in the CTS100 scenario is higher than in the CTS60 scenario, export revenues in this scenario decrease more. On the other hand, despite a significant reduction in investment costs in the HPF and HVCF scenarios, the costs of these scenarios have not changed much compared to the BAU scenario. This is because the model, in order to earn more revenue from energy exports, especially hydrogen exports, has built more capacity for technologies, which has led to a significant increase in energy exports, especially hydrogen. On the other hand, in the unsubsidized energy prices scenario, both system costs have increased, and export revenues have decreased. It is important to note that Iran pays a large amount of subsidies for energy carriers to its people every year [89]. These subsidies make up a large portion of the country's budget. According to the national energy subsidy reform [91]. if the prices of energy carriers are deregulated and brought closer to world market prices, although the energy sector will face increased costs and reduced exports, the saved financial resources that had to be paid for

Table 7
Comparison of scenarios based on three important indicators defined in the two sections. The values are in % and if the scenario's performance has improved, it is shown in green, and if it has worsened, it is shown in red.

	GHG emissions		Cost		Export income	
	Total energy supply system	Hydrogen supply chain	Total energy supply system	Hydrogen supply chain	Total energy supply system	Hydrogen supply chain
CTS10	-6.38	-12.57	0.56	5.68	-0.58	-10.38
CTS30	-13.45	-33.11	2.26	13.92	-1.53	-15.36
CTS60	-41.88	-72.48	-4.16	19.93	-6.67	-18.83
CTS100	-46.31	-88.78	0.73	22.06	-6.05	-19.37
HPF	23.31	40.19	0.92	-0.51	0.62	3.64
HVCF	7.361	-27.28	1.23	-1.94	0.70	45.27
ECS40	-21.47	-53.47	-4.15	7.05	-2.59	-19.28
EDS	-1.73	-4.13	57.98	15.74	-8.78	-19.24

energy subsidies can be reimbursed for developing infrastructure, creating jobs, investing in education, health, and other sectors in the country, and overall bringing significant economic and social growth to the country. Also, this capital can help in the further development of renewable systems and accelerate the energy transition process in Iran. For example, given Iran's high potential for using solar systems [66], part of the cost that was to be spent on energy subsidies can be allocated to this sector and the capacity of these technologies can be increased, then electricity production and exports would increase, and in the long term, electricity export revenues would increase compared to the business-as-usual scenario. However, this issue requires more detailed analysis and consideration of all sectors of the economy along with the energy sector and depends on various indicators and policies.

In addition to the above, comprehensive policy analyses can be extracted from the model results. First, it can be said that the model prioritizes the development of hydrogen production technologies based on gas over other technologies. This indicates that hydrogen production from natural gas is much more cost-effective and efficient in Iran than from electricity. This is especially important given that a large portion of the country's electricity is generated from natural gas. On the other hand, Iran, with its abundant gas resources, very low levelized costs of fossil fuel energy carriers, and the collection of a large volume of flared gas, has a unique opportunity to produce hydrogen from natural gas using various technologies. It can gain a significant competitive advantage by pricing hydrogen lower than other hydrogen-producing countries. The existence of large refineries in Iran also creates the opportunity to supply part of the country's hydrogen demand as a refinery by-product.

Secondly, producing electricity from fuel cell technology on a large scale isn't cost-effective under current cost conditions, so the model doesn't choose this technology. It's important to note that fuel cell technology has only been included in the model as an option for electricity generation. Also, the energy security benefits of distributed power generation have not been considered in the paper. If fuel cell technology were considered as an energy storage device for renewable systems, the model may show different results.

Additionally, in all the scenarios considered, hydrogen is not produced from grid electricity. This indicates that producing hydrogen from gas is much more cost-effective than from electricity, Therefore, the model decides to meet the hydrogen demand from gas as much as possible and export any surplus electricity.

Finally, if the primary focus is on the hydrogen supply chain within the country, and all efforts and planning are directed toward expanding this energy carrier, then providing incentives to the entire hydrogen supply chain (HVCF) would be the most effective approach overall, considering all the mentioned indicators. In this scenario, the best technology for hydrogen production would be electrolyzers powered by renewable and solar energy systems.

It's worth noting that the results of this study align with other research. For example, according to Table 1, most studies on Iran's energy supply system have identified combined-cycle power plants as the most suitable technology for meeting long-term electricity demand. The results of the current study confirm this, as Fig. 7 shows that combined-cycle will supply over 60% of annual electricity demand by 2055. Additionally, many previous studies have highlighted Iran's high potential for producing hydrogen using renewable technologies [49,50], a finding that this research also clearly demonstrated this issue in the HVCF scenario.

5. Conclusions

In this research, comprehensive and detailed modeling and optimization of the country's energy supply system was conducted with the aim of achieving sustainable development and increasing hydrogen penetration. The developed model in this research can be used as a scientific basis for optimal policymaking for the development of the energy supply system by 2055 and providing a comprehensive national roadmap for hydrogen supply. To this end, the energy supply system was modeled using the OSeMOSYS model. For this purpose, technical data (efficiency, lifespan, capacity factor, availability factor, etc.) and economic data (fixed cost, variable cost, investment cost, etc.) of all energy production, extraction, conversion, transmission, and distribution technologies were collected and then analyzed. The results show that if the current situation continues, a significant portion of the country's energy supply system burden will be on natural gas. Also, the most suitable technology for electricity generation is the combined cycle, supplying up to 54.161% of the total electricity demand. Following the validation and verification of the developed energy supply model, the hydrogen supply chain was added to the model, and the model was evaluated under nine different scenarios. The results show that in the BAU scenario, SMR technology will be the best option for hydrogen production, requiring an investment of USD 9.5 billion. However, this approach may limit environmental benefits. Subsequently, the different scenarios were compared in terms of environmental and economic indicators, showing the impact of each scenario on the hydrogen supply chain and the entire energy supply system of the country. For example, we further observed that imposing carbon taxes incentivizes cleaner hydrogen technologies. Higher tax rates (e.g., $60-$100 per ton) drive transitions to low-carbon

technologies like pyrolysis, and 40% emission reduction target significantly shifts hydrogen production to cleaner technologies like blue and turquoise hydrogen. On the other hand, providing financial incentives for hydrogen production and the entire supply chain (HPF and HVCF) enhances hydrogen export potential and can position Iran as a key player in the hydrogen export market. However, this may increase emissions unless aligned with renewable technologies. Therefore, based on the results obtained (Table 6), if the primary focus is on the growth of the hydrogen supply chain, the HVCF scenario represents the most effective method for hydrogen production. In this scenario, hydrogen is generated using electrolyzer technology that is powered by renewable energy sources. Also, considering the decrease in natural gas production in the long term, electrolyzer technology combined with solar power can be a suitable alternative for hydrogen production. In the following, gradual removal of energy subsidies reduces total hydrogen production from natural gas, demonstrating the reliance on subsidized fossil fuels.

5.1. Policy implications

Iran is among the top five countries in the world with natural gas reserves. But if no action is taken to develop gas fields or install and start up compressors to increase the pressure of fields and transmission and distribution lines, considering the increasing demand, Iran will face a severe gas shortage in the long term. This issue is of greater importance considering that the main load of Iran's energy supply system is based on natural gas (Fig. 1).

On the other hand, given the dwindling fossil fuel reserves in Iran, the capacity of renewable energy systems is expected to grow. Nevertheless, the adoption of low-carbon scenarios can expedite this energy transition and minimize its corresponding economic and social costs.

In the end, it can be said that scenarios with strict emission controls or high carbon taxes achieve substantial emission reductions but at higher economic costs or reduced export revenue. Aligning policies with global market trends, such as desubsidizing fossil fuels, can promote renewable energy adoption but requires comprehensive economic planning. Therefore, following desubsidization targets may result in higher costs for hydrogen, which needs international support.

5.2. Limitations and future research

This research is the first large-scale energy supply model in Iran that incorporates the hydrogen supply chain. Throughout the modeling process, we encountered various limitations and identified opportunities for further research based on our current work. Firstly, due to existing limitations, such as the diversity of technologies, particularly in the oil and gas sectors, and the aging of refinery technologies, most data for this model was collected from local sources. Secondly, water is closely related to the energy supply system and the hydrogen supply network. Additionally, not accounting for land constraints in technology construction and adopting a single-region approach have limited our ability to identify opportunities and challenges in the energy sector. Expanding to a multi-regional assessment and implementing an integrated energy, water, and land model at a national level can provide deeper insights into the exact location of technology construction and energy carrier transportation, water accessibility, suitable locations for renewable energy technologies, locations for injecting captured CO_2 into the ground and utilizing it, and so on. Thirdly, our study did not examine the macroeconomic impacts of the energy sector on other non-energy sectors, such as spending carbon taxes collected for infrastructure development, the amount of employment created, the amount of liquidity generated from providing subsidies for hydrogen technologies, and economic growth. By considering these factors, a more comprehensive roadmap for the future of the energy supply system and increasing hydrogen penetration can be drawn. Fourthly, in this study, only pure hydrogen production technologies were evaluated, and the opportunities to use hydrogen produced from refinery by-products as an energy carrier were not examined. Fifthly, based on the literature review conducted, each of the scenarios that were investigated is among the most well-known pathways that various countries strive to implement as policies or laws to achieve a sustainable, low-carbon energy system. Therefore, it was important to analyze the impact of each scenario on the energy supply system individually. These findings allowed us to identify the specific effects of each policy on different sectors of Iran's energy system, particularly regarding the increased penetration of the hydrogen supply chain. Nevertheless, the inclusion of combined scenarios that incorporate two or more of the aforementioned policies could yield novel and innovative insights, ultimately assisting in the formulation of a more precise and comprehensive roadmap. Finally, similar to many macroeconomic modeling and optimization analyses, energy demand was entered as an exogenous input in the form of final energy, and the impact of the mentioned scenarios on energy demand was not evaluated.

Nevertheless, this study represents an initial step toward identifying opportunities and challenges associated with this field in Iran. In the future works, by using open-modeling infrastructures and the data available in this study and considering more accurate and detailed assumptions, scenarios, and technologies and adding them to this model, the quality of the insights obtained can be further improved.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Most of the data used in the modeling are open to be provided upon request. Some pieces of data on sectoral demand and resource availability are confidential and need permission to be shared.

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